

the reverse biased direction, there is no voltage output across resistor R . A1

- 7 (a) Electromotive force is the work done in transforming non-electrical energy into electrical energy per unit charge passing through the terminals of the source. B1
Potential difference is the amount of electrical energy transformed per unit charge to some other forms of energy when the charge passes from one point to the other. B1

- (b) (i) Since $I = I_1 + I_2$,

$$I = \frac{V}{R} + \frac{V}{R_V} \quad \text{M1}$$

$$\frac{1}{R} = \frac{I}{V} - \frac{1}{R_V}$$

$$R = \frac{R_V V}{IR_V - V} \quad \text{A1}$$

(ii) $V = IR + IR_A$ M1

$$\frac{V}{I} = R + R_A$$

$$R = \frac{V}{I} - R_A$$
 A1

(iii) For $R = \frac{V}{I}$, $R_V \gg R$ and $R_A \ll R$. Hence, R_V should be infinite, A1

and R_A should be equal to zero. A1

(c) (i) $R = \rho \frac{l}{A}$

$$R = 1.4 \times 10^{-6} \times \frac{1.1}{\pi(0.304 \times 10^{-3})^2}$$
 M1

$$R = 5.30 \, \Omega$$
 A1

(ii) 1. $\frac{l}{110} \times \frac{5.3}{5.3 + 0.30 + 0.50} \times 2.2 = \frac{1.2}{1.2 + 1.1} \times 1.8$ M1

$$l = 54.0 \text{ cm}$$
 A1

2. E.m.f. of cell is larger than the terminal p.d. of cell
length AC will increase. M1
A1

(iii) 1. $I = \frac{V}{R} = \frac{2.2}{(0.3 + 5.3)}$ M1
 $= 0.3929 \text{ A}$

$$P = I^2 R = (0.3929)^2 (5.30) = 0.818 \text{ W}$$
 A1

2. $I = Anv$

$$0.3929 = \pi(0.304 \times 10^{-3})^2 (10^{29}) v (1.6 \times 10^{-19})$$

$$v = 8.46 \times 10^{-5} \text{ m s}^{-1}$$
 A1

(d) (i) $R = \frac{3.4}{1.125} = 3.02 \, \Omega$ A1

(ii) As the potential difference (p.d.) increases from 0 V to 3.4 V, the ratio of V to I decreases, hence resistance decreases. B1
As the p.d. increases from 3.4 V to 6.0 V, the ratio of V to I increases. Hence, resistance increases during this interval. B1